CSE 655 — Network Science

Assignment 2 Report

Winter 2026

This report summarises the results and observations from all five questions of Assignment 2.

Datasets used:

- Q1, Q2, Q3: Algorithmically generated networks (no external data)
- Q4: Barabási-Albert, Erdős–Rényi, and email-Eu-core (SNAP)
- Q5: Yeast Protein Interactome — LC-multiple dataset (CCSB Harvard) + SGD Phenotype data

Question 1: Watts–Strogatz Small-World Network Model

Parameters: $n = 1000$, $k = 10$, rewiring probability p from 10^{-4} to 1 (log scale, 20 points, averaged over 10 runs).

The Watts–Strogatz model generates networks by starting from a regular ring lattice and rewiring each edge with probability p. We track how the scaled clustering coefficient $C(p)/C(0)$ and scaled path length $L(p)/L(0)$ change as p increases.

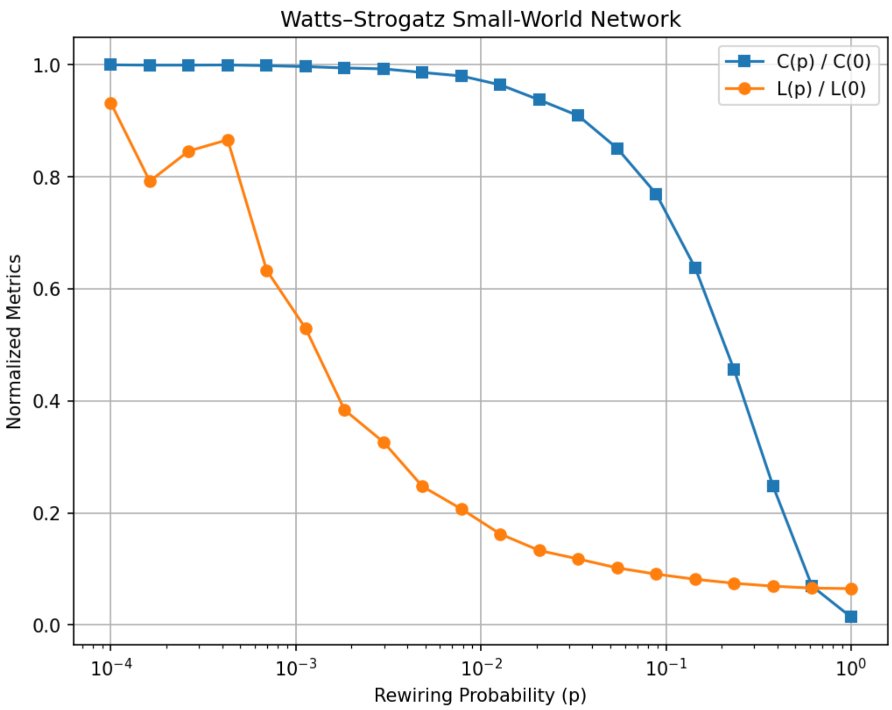


Figure 1: Scaled $C(p)/C(0)$ and $L(p)/L(0)$ vs rewiring probability p.

The plot replicates the classic Watts–Strogatz result. $L(p)/L(0)$ drops sharply even at small p — a few random shortcuts are enough to drastically reduce average path length. $C(p)/C(0)$ stays high until around $p \approx 0.1$, where most edges are still local. The small-world regime lies in the range $0.01 p 0.1$, where both properties coexist.

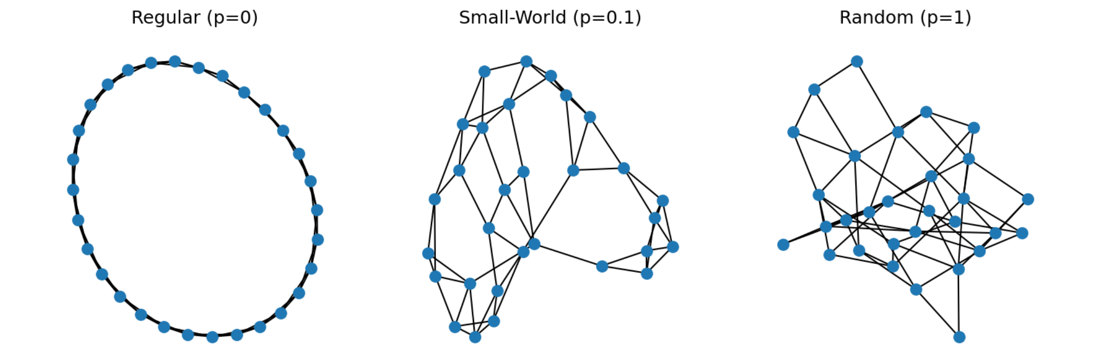


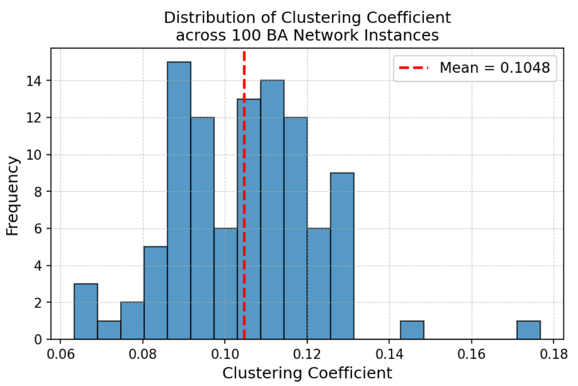
Figure 2: Network structure at $p = 0$ (regular), $p = 0.1$ (small-world), $p = 1$ (random).

The three network layouts illustrate the structural transition. The regular lattice has a clear ring structure; the small-world network has a few long-range shortcuts while retaining local clusters; the random network has no visible structure.

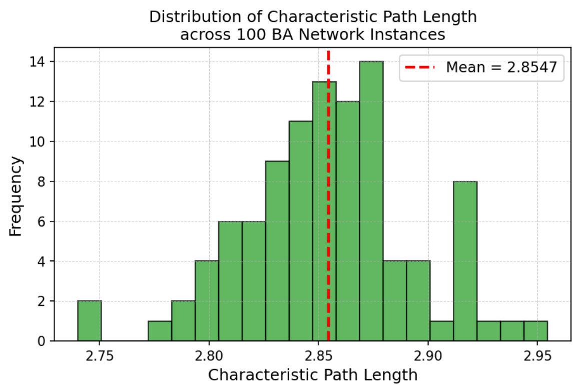
Question 2: Barabási-Albert Scale-Free Network

Parameters: $n = 200$, $m_0 = 5$ (initial clique), $m = 3$ (edges per new node). Metrics averaged over **100 independent instances**.

The BA algorithm grows a network by adding nodes one at a time, each connecting to existing nodes with probability proportional to their degree (preferential attachment). The entire algorithm was implemented from scratch.



(a) Distribution of clustering coefficient across 100 BA instances.



(b) Distribution of characteristic path length across 100 BA instances.

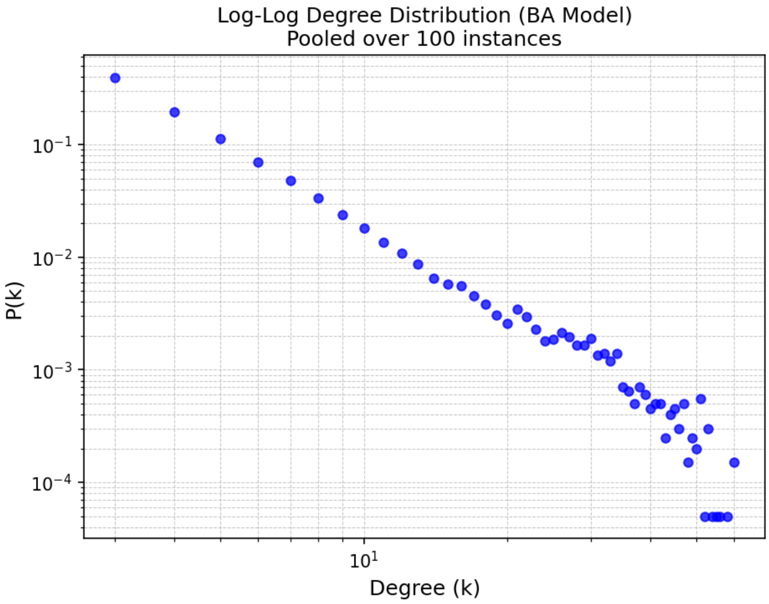


Figure 4: Log-log degree distribution pooled over 100 BA instances.

The log-log degree distribution is approximately linear, confirming the power-law behaviour $P(k) \propto k^{-3}$ predicted by the BA model. The clustering coefficient is consistently low across instances (typically $\langle C \rangle \approx 0.10$–$0.13$) — a known characteristic of the basic BA model where connections are driven by degree rather than local clustering. The characteristic path length is short ($L \approx 2.8$), reflecting the small-world property of scale-free networks.

Question 3: Modified BA Algorithm (k^α Preferential Attachment)

Parameters: $n = 300$, $m_0 = 5$, $m = 3$, 20 instances per variant. Exponent $\alpha \in \{1, 2, 3, 4\}$.

We modify the attachment probability to $\Pi(k_i) \propto k_i^\alpha / \sum_j k_j^\alpha$. As α increases beyond 1, the rich-get-richer effect is amplified — high-degree nodes receive even more connections relative to low-degree ones.

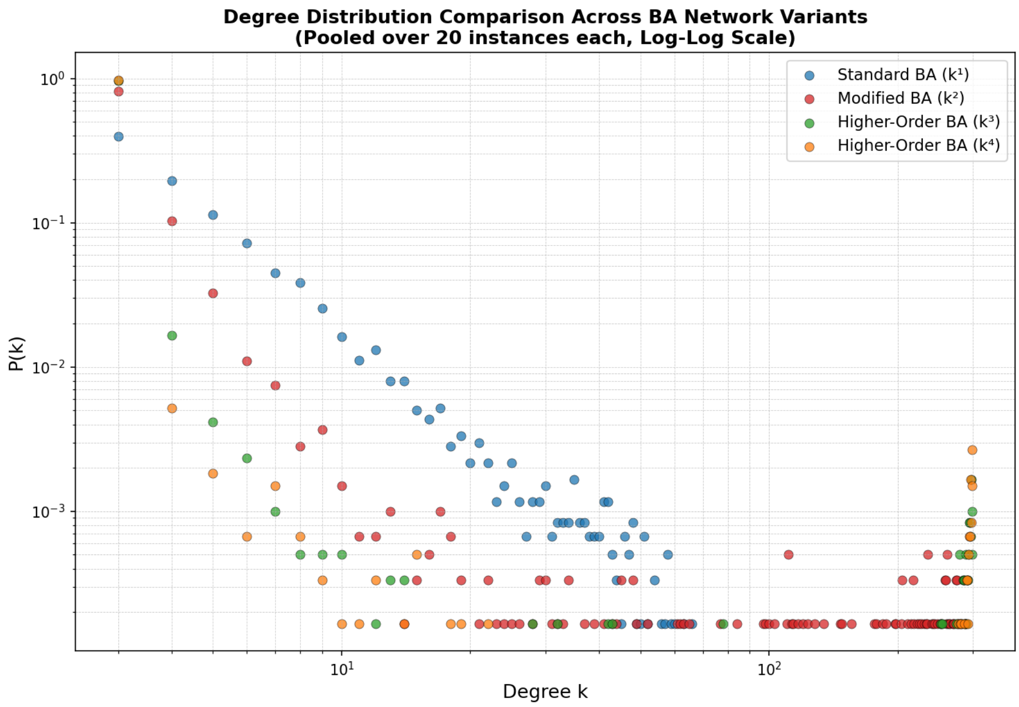
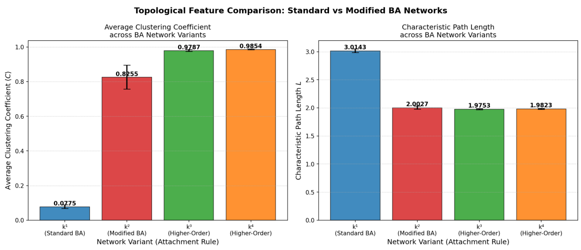
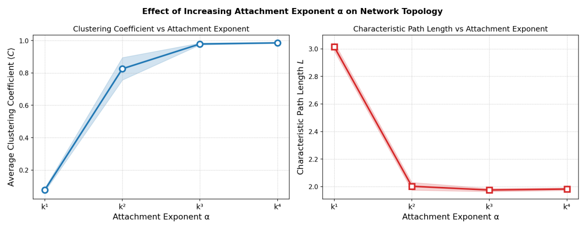


Figure 5: Degree distribution for all four α variants (log-log, overlaid).



(a) Average clustering coefficient and path length across variants.



(b) Topological metrics as a function of α.

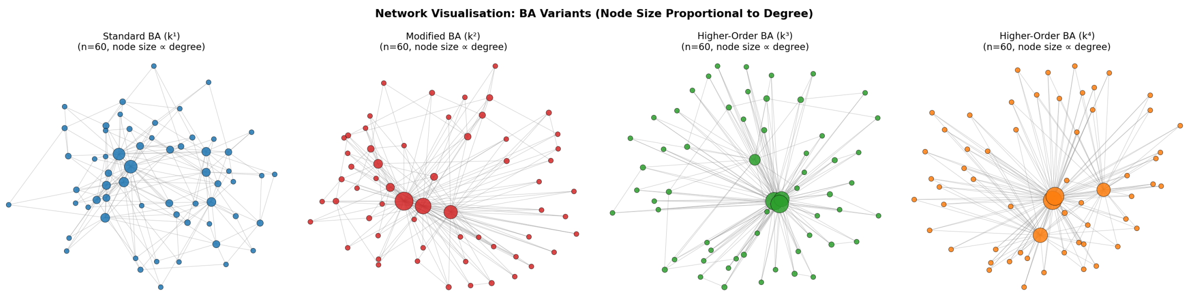


Figure 7: Network structure for each variant ($n = 60$, node size proportional to degree).

As α increases, the degree distribution tail becomes steeper — hub nodes accumulate an even greater share of connections. With $\alpha = 4$, the network approaches a star-like structure with one dominant hub. Clustering coefficient decreases with α since new nodes preferentially connect to the hub rather than forming local triangles. The network visualisations confirm this: at k^4, one large hub is clearly visible with most nodes as peripheral spokes.

Question 4: Network Robustness — Node Deletion

We study how networks respond to node removal by tracking giant cluster size S, average fragment size $\langle s \rangle$, and characteristic path length L. Two strategies: **random failure** (random removal) and **targeted attack** (highest-degree nodes first).

Networks tested: Barabási-Albert (SF, $n = 1000$, $m = 3$), Erdős–Rényi (same n and average degree), and the email-Eu-core real-world network (1005 nodes).

Part (a): Random Deletion on BA Network

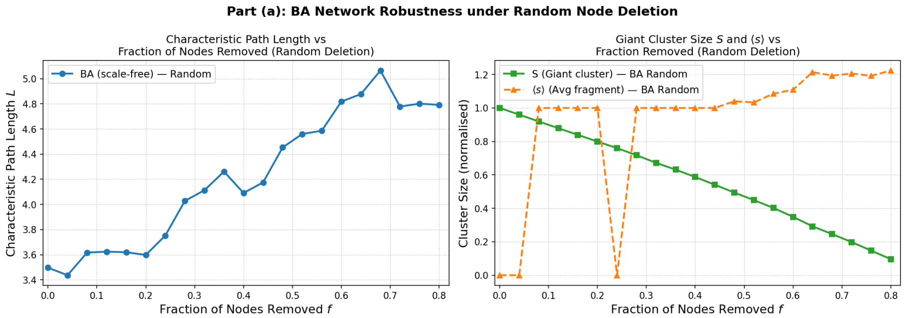


Figure 8: $L(f)$ and $S(f)$ under random deletion on the BA network.

Under random deletion, the BA network is highly robust — S decreases slowly even as f grows. Most randomly removed nodes are peripheral low-degree nodes that don't significantly affect global connectivity.

Part (b): Random vs Scale-Free Networks

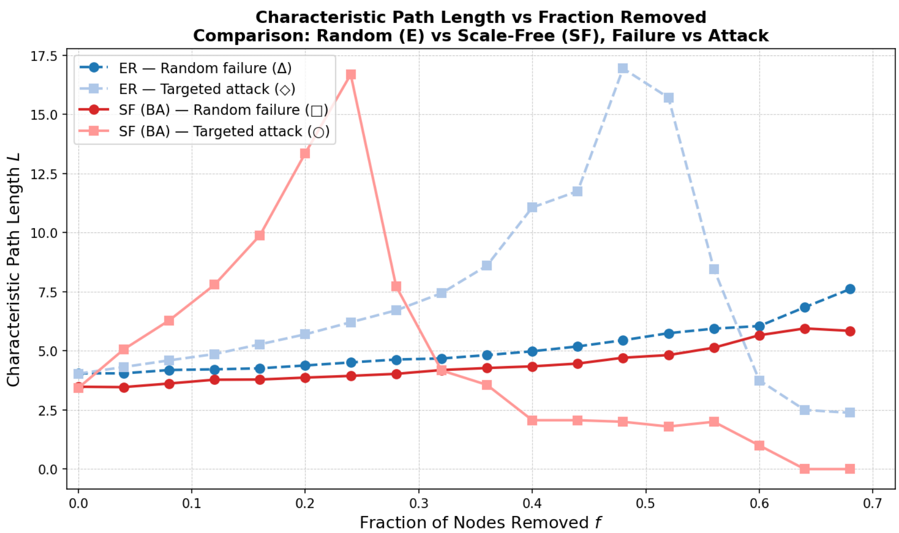


Figure 9: Characteristic path length $L(f)$ — ER vs SF, random vs targeted.

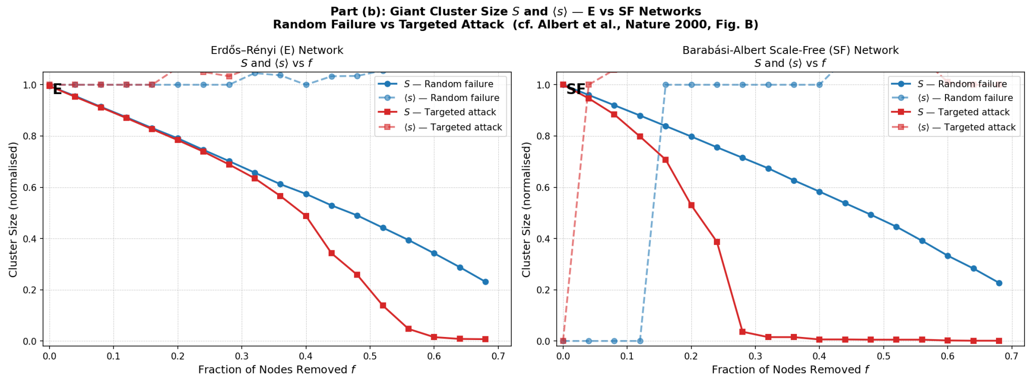


Figure 10: S and $\langle s \rangle$ — ER vs SF, random vs targeted (replicating Albert et al. Fig. B).

The ER network shows similar responses to both strategies — no structural asymmetry because the degree distribution is homogeneous. The SF (BA) network shows a clear asymmetry: robust under random failure (S drops slowly), but fragile under targeted attack (S drops sharply). This matches the expected pattern from Albert et al. (2000).

Part (c): Real-World Network (email-Eu-core)

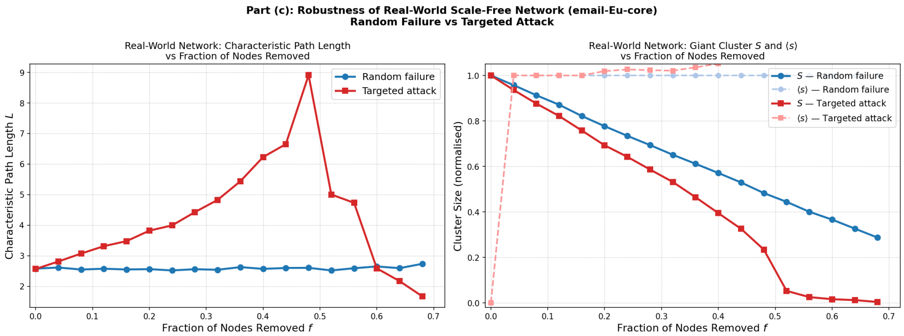


Figure 11: Robustness of email-Eu-core network under random failure and targeted attack.

The email-Eu-core network (1005 nodes) behaves like a scale-free network — slow S decay under random failure, rapid decay under attack on high-degree nodes. This is expected since email networks follow a heavy-tailed degree distribution.

Part (d): Summary

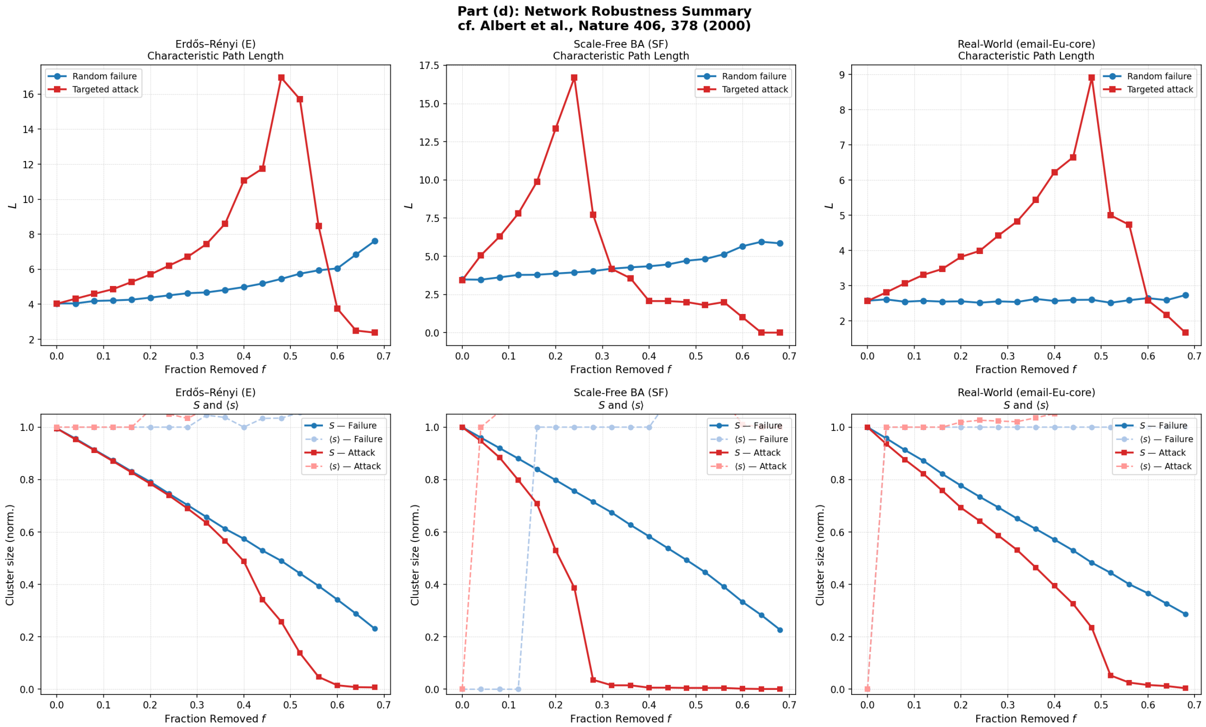


Figure 12: Full summary: L and S across all three networks and both strategies.

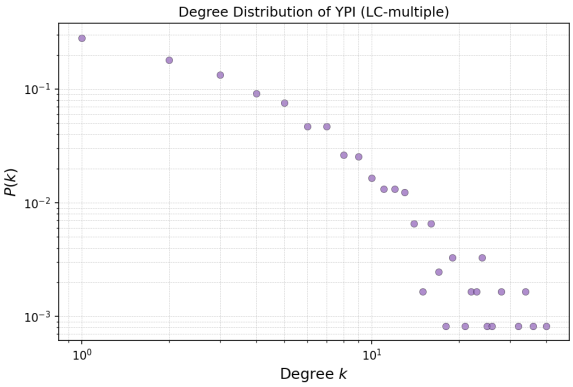
Our results are consistent with Albert et al. (2000). Scale-free networks are simultaneously robust to random failures and fragile to targeted attacks — a direct consequence of their heterogeneous degree distribution. ER networks lack this asymmetry. The percolation threshold f_c under targeted attack on SF networks is noticeably lower than under random failure, matching the reference findings.

Minor differences from the reference (smoother curves, exact f_c values) are expected since we use smaller networks ($n = 1000$ vs ~ 10000) and sample-based path length estimation.

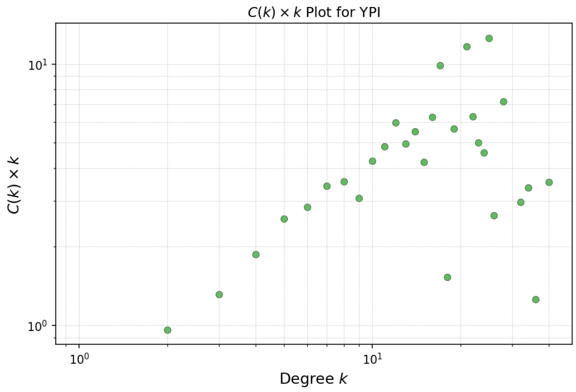
Question 5: Yeast Protein Interactome (YPI)

Dataset: **LC-multiple** from the CCSB Yeast Interactome Database — 2924 literature-curated protein interactions. Essential gene annotations from the SGD Phenotype Database (null mutants with "inviable" phenotype).

Part (a): Network Properties



(a) Degree distribution of YPI (log-log scale).



(b) $C(k) \times k$ plot for YPI.

The YPI degree distribution is heavy-tailed on the log-log scale, consistent with scale-free behaviour. A small number of hub proteins have many interactions while most proteins have only a few.

The $C(k) \times k$ plot shows that low-degree proteins tend to participate in tightly clustered local modules, while high-degree hub proteins bridge between them with lower local clustering. This hierarchical modular structure is characteristic of biological protein networks.

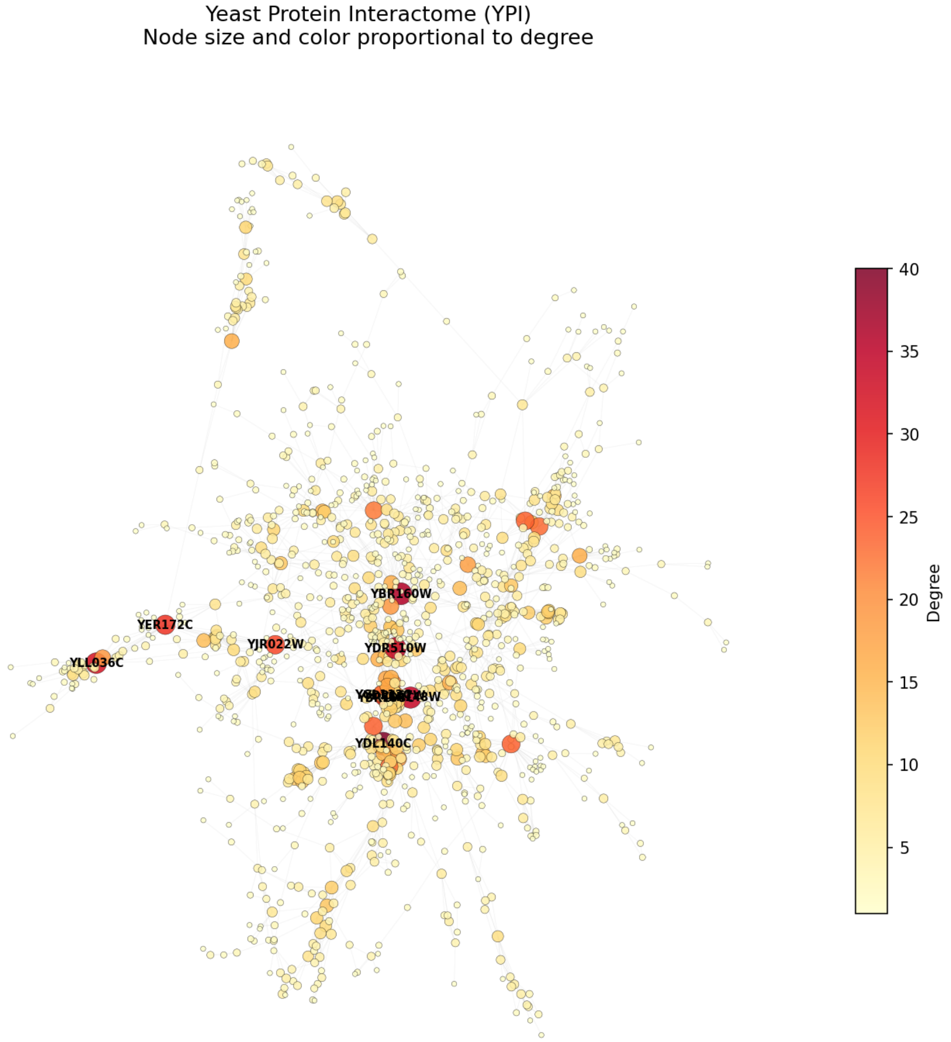


Figure 14: YPI network visualisation — node size and colour proportional to degree.

The network visualisation highlights hub proteins (large, dark-red nodes) vs peripheral proteins (small, yellow nodes). The top 10 highest-degree proteins are labelled. The Cytoscape visualisation (imported from `ypi_edges.csv` and `ypi_nodes.csv`) provides an interactive version with degree-based styling.

Figure 15: YPI network in Cytoscape — node size and colour proportional to degree.

Part (b): Key Proteins and Essential Genes

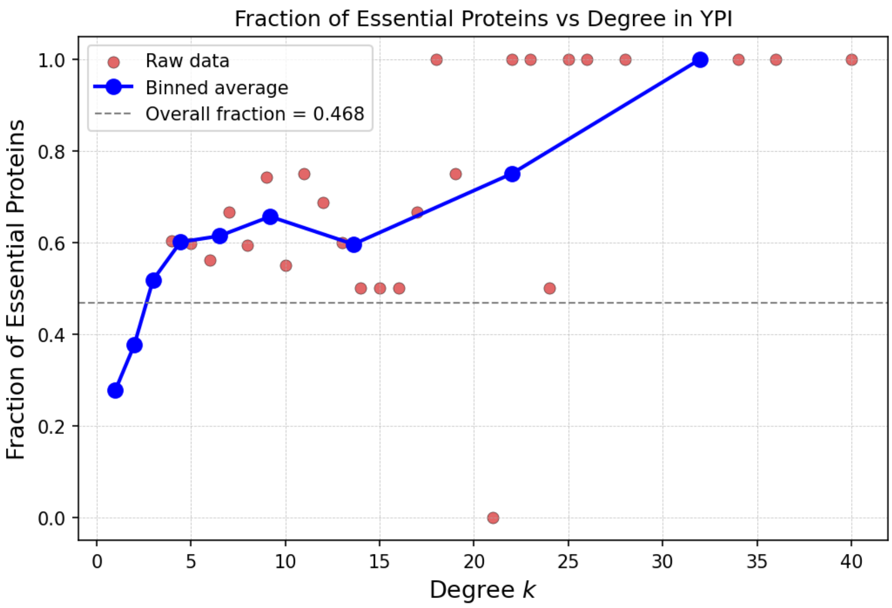


Figure 16: Fraction of essential proteins vs degree k in YPI.

High-degree proteins are more likely to be essential — their deletion is lethal to the cell. This is the **centrality-lethality rule** (Jeong et al., *Nature* 2001). The trend in the binned average clearly shows that as degree increases above the network average, the fraction of essential proteins increases. Hub proteins are involved in multiple pathways, so losing them disrupts many cellular functions simultaneously.

Part (c): Robustness of YPI

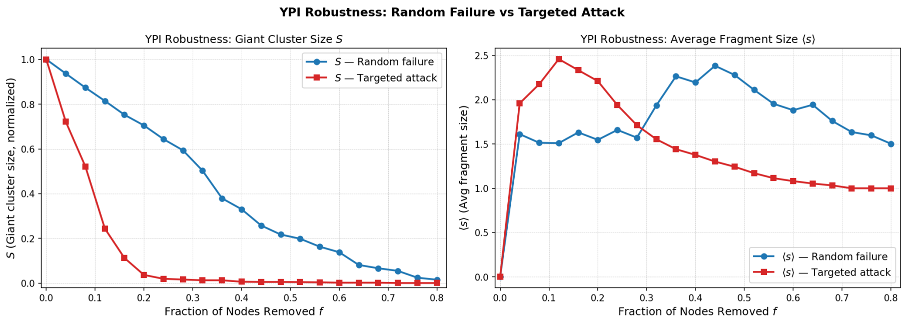


Figure 17: S under random failure vs targeted attack on YPI.

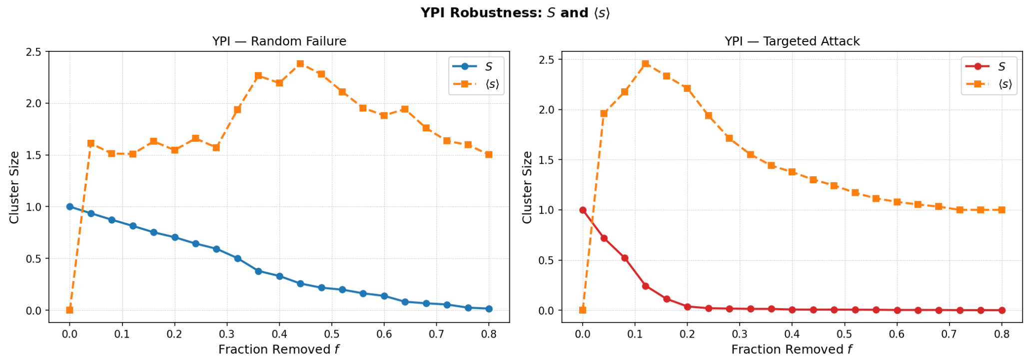


Figure 18: S and $\langle s \rangle$ combined for each strategy on YPI.

The YPI shows the same robustness asymmetry as other scale-free networks (Q4): slow decline in S under random failure, but rapid fragmentation under targeted attack. $\langle s \rangle$ peaks near the percolation threshold under attack, indicating rapid breakup into small fragments.

Crucially, the proteins removed first in a targeted attack (highest-degree hubs) are the same ones identified as essential in Part (b). This links network topology directly to biological essentiality — the most connected proteins are both structurally critical and biologically indispensable.

Conclusions

1. **Small-world networks** (Q1) exist in a narrow parameter range between regular and random, with both high clustering and short paths — well captured by the WS model.
2. **Scale-free networks** (Q2, Q3) arise from preferential attachment and exhibit power-law degree distributions. Amplifying this mechanism (k^α, $\alpha > 1$) leads to more extreme hub dominance and reduced clustering.
3. **Network robustness** (Q4, Q5) is fundamentally shaped by topology. Scale-free networks are robust to random failures but fragile to targeted attacks — a direct result of their heterogeneous degree distribution. This pattern holds in both synthetic and real-world networks.
4. In the **yeast interactome** (Q5), network structure reflects biology: hub proteins are more likely to be essential, and targeted removal of hubs is biologically equivalent to removing essential genes.